

MINI PROJECT 2: SSH BRUTE FORCE SIMULATION

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1. Incident Summary

This report documents a simulated SSH Brute Force attack against an internal production server. The attack sequence involved a high volume of unauthorized login attempts, eventually resulting in a successful compromise. The investigation details the source of the attack, its classification, and essential forensic response steps.

2. Attack Telemetry Details

Attribute	Value
Attack Vector	SSH Brute Force + Unauthorized Access
Attacker Source IP	45.33.32.156 (Public Network)
Destination Internal IP	192.168.1.20 (Production Server)
Service Targeted	Secure Shell (SSH) — Port 22
Incident Timeline	Start: 02:14 AM Breach: 02:17 AM
Brute Force Metric	50 Login Failures within 180 seconds

3. Severity & Risk Matrix

Based on the successful unauthorized login, this incident is classified as **CRITICAL**.

Risk Factor	Forensic Observation	Risk Rating
Attacker Origin	Public IP Address identified	HIGH
Compromise Status	Active unauthorized session confirmed	CRITICAL
Temporal Context	Post-business hours (02:17 AM)	HIGH
Asset Impact	Internal high-value production server	HIGH

4. Immediate Forensic Response

- **PERIMETER BLOCK:** Null-route source IP 45.33.32.156 at the firewall.
- **NETWORK ISOLATION:** Quarantining server 192.168.1.20 to stop potential lateral movement.
- **ACCOUNT AUDIT:** Identifying the specific user account utilized during the breach.
- **CREDENTIAL RESET:** Forcing a global password reset for the affected user.
- **POST-BREACH ANALYSIS:** Reviewing shell history for suspicious command execution.
- **EVIDENCE RETENTION:** Snapshotting system logs for formal digital forensics reporting.

5. Operational Escalation

Field	Escalation Details
Escalation Path	L2 SOC Analyst / Incident Response Team
Incident Priority	P1 — Critical Breach
Justification	Unauthorized internal access achieved via external IP

6. Strategic Recommendations

- Deploy automated defense tools like **fail2ban** to throttle repeated login failures.
- Enforce **Public Key Authentication** and disable password-based SSH access.
- Restrict management access using **Trusted IP Whitelists**.
- Implement **MFA (Multi-Factor Authentication)** for all server administrative tasks.
- Configure SIEM thresholds for immediate alerting on brute-force signatures.

7. Conclusion

The simulation confirms that a weak authentication policy allowed for a successful SSH Brute Force attack. The breach emphasizes the need for robust identity management and real-time network monitoring. Immediate server hardening and credential rotation are required to mitigate further risk.